Name: ____________

Date: ____________

# Quadratic Functions & Equations

## Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce | D — Deepen | A — Apply | R — Reinforce

☐ **1. I can identify key features of a quadratic function from its graph or equation.**

RELATED SKILLS

- I — Determine the vertex and axis of symmetry.
- I — Determine x- and y-intercepts of a quadratic function.
- I — Identify standard, factored, and vertex forms.

☐ **2. I can connect standard, factored, and vertex forms to features of a parabola.**

RELATED SKILLS

- I — Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
- D — Connect parameters in a formula to features of its graph.
- D — Identify standard, factored, and vertex forms.
- A — Determine the vertex and axis of symmetry.
- A — Determine x- and y-intercepts of a quadratic function.

☐ **3. I can solve a quadratic equation by factoring.**

RELATED SKILLS

- I — Determine real and complex zeros using an appropriate method.
- D — Apply the zero-product property.
- A — Factor a monic quadratic trinomial.
- A — Factor a quadratic trinomial with a nonunit leading coefficient.

☐ **4. I can solve a quadratic equation using the square-root property.**

RELATED SKILLS

- I — Apply the square-root property with both signs.
- A — Extract perfect-power factors from a radical.

☐ **5. I can solve a quadratic equation by completing the square.**

RELATED SKILLS

- I — Create a perfect-square trinomial by completing the square.
- A — Simplify an algebraic expression using operation properties.

☐ **6. I can solve a quadratic equation using the quadratic formula.**

RELATED SKILLS

- I — Substitute coefficients accurately into the quadratic formula.
- I — Use the discriminant to predict the number and type of solutions.
- A — Extract perfect-power factors from a radical.

☐ **7. I can choose an efficient method for solving a quadratic equation.**

RELATED SKILLS

- I — Select an efficient quadratic-solving method from equation structure.
- A — Apply the square-root property with both signs.
- A — Apply the zero-product property.
- A — Create a perfect-square trinomial by completing the square.
- A — Substitute coefficients accurately into the quadratic formula.

☐ **8. I can graph a quadratic function using its vertex, intercepts, and symmetry.**

RELATED SKILLS

- D — Connect parameters in a formula to features of its graph.
- D — Determine the vertex and axis of symmetry.
- D — Determine x- and y-intercepts of a quadratic function.
- A — Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.

☐ **9. I can create and interpret a quadratic model for a projectile, area, or optimization problem.**

RELATED SKILLS

- I — Interpret vertex and zeros in an application.
- D — Explain a model parameter in the language of its context.
- A — Attach and interpret units for rates, inputs, and outputs.
- A — Restrict a model to meaningful contextual inputs.